

SENTRAIS OS ARCHITECTURE

Six-Layer Intelligence Stack Powering High-Stakes Operations

LAYER 1 INGESTION

Data Acquisition & Real-Time Streams

Sensors, IoT, SCADA/OT systems, CAD dispatch, weather feeds, ticketing, access control, broadcast signals. Unified ingestion pipeline handling 10M+ events/second from venues, cities, and partner systems.

LAYER 2 INTELLIGENCE

SIPE Engine (Sentrais Intelligence Processing Engine)

ML models for anomaly detection, predictive analytics, natural language processing. Pattern recognition across cyber-physical domains. Gemini/Vertex AI integration for real-time inference and report generation.

LAYER 3 EVIDENCE

Evidence Ledger & Compliance Repository

Immutable audit trail of all decisions, actions, and system states. BigQuery-backed analytics. Automated compliance reporting for FEMA, insurance, and regulatory requirements. AAR/IP auto-generation.

LAYER 4 ORCHESTRATION

Dual-Engine Orchestration (Resilience + Innovation)

Digital playbook execution, workflow automation, multi-agency coordination, real-time alerting. Two parallel engines: Resilience Orchestration (operations) and Innovation Orchestration (capability building).

LAYER 5 APPLICATION

Product Surfaces & User Interfaces

EVERGAME (sports), eVenu (venues), CiviGrid (cities), Federal (DHS/FEMA). Command dashboards, mobile apps, embedded analytics, partner APIs. Role-based access across all stakeholder types.

LAYER 6 SECURITY

Zero-Trust Security & Compliance Framework

Chronicle Security for cyber-physical fusion. FedRAMP High pathway. Five controlled access zones (Core, Playbook, Client, Integration, Agent). End-to-end encryption, SOC 2 Type II, NIST 800-53 alignment.

THE SENTRAIS DIFFERENCE: Unlike point solutions that address single layers, Sentrais OS provides a unified intelligence stack where each layer feeds the others. Data flows up through ingestion and intelligence to inform orchestration. Evidence flows down to validate decisions. This creates compounding intelligence—every deployment makes the entire system smarter.

DUAL ORCHESTRATION ENGINES

Two Engines. One Platform. Unified Evidence. Continuous Learning.

RESILIENCE ORCHESTRATION

Protecting Operations Under Pressure

INNOVATION ORCHESTRATION

Accelerating Capability Building

1

READINESS & COMPLIANCE

Living blueprints, certification pre-checks, digital simulations, compliance validation, risk scoring

1

DISCOVERY & VALIDATION

Innovation testing, POC simulations, multi-stakeholder validation, vendor assessment, feasibility scoring

2

LIVE OPERATIONS

Run the playbook, real-time response, multi-agency COP, incident management, dynamic resource allocation

2

PILOT OPERATIONS

Controlled deployment, real-time measurement, coordinated trials, A/B testing, performance tracking

3

AFTER-ACTION LEARNING

Evidence ledger analysis, auto-generated analytics, benchmarking, gap identification, improvement planning

3

LEARNING & INTEGRATION

Evidence-based assessment, cross-venue learning, benchmarks, ROI analysis, integration roadmaps

4

CERTIFICATION & RENEWAL

Readiness scores, insurance alignment, regulatory compliance, continuous certification, renewal automation

4

SCALE & CERTIFICATION

Innovation credentials, vendor certification, continuous cycles, portfolio rollout, ecosystem expansion

SENTRAIS OS CORE

Evidence Ledger (immutable audit trail) | **SIPE Engine** (ML/AI processing) | **NIN Methodology** (5D framework)

RESILIENCE protects what you have.

Game day safety, incident response, regulatory compliance, operational continuity.

INNOVATION builds what you need.

New technology adoption, vendor coordination, capability expansion, competitive advantage.

SENTRAIS CORPORATION | The Intelligence Operating System for High-Stakes Environments | sentrais.com